

BENTWATERS, United KingdomWMO: **035963**Lat: **52.133N**Lon: **1.433E**Elev: **26**StdP: **101.01**Time Zone: **0.00 (EUW)**Period: **82-92**WBAN: **35048****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.1	-2.0	-8.9	1.8	-2.0	-7.1	2.1	-1.2	15.2	9.3	13.5	8.2	2.6	90	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	24.8	18.1	23.2	17.4	22.0	16.8	19.2	23.6	18.3	22.1	17.5	20.9	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.4	20.6	16.8	12.0	20.2	16.0	11.4	19.6	54.4	23.3	51.6	22.0	49.3	21.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.1	8.0	-5.9	28.0	4.1	2.8	-8.8	30.0	-11.3	31.6	-13.6	33.2	-16.6	35.2		
			-6.9	20.8	3.9	0.9	-9.7	21.5	-12.0	22.0	-14.1	22.5	-16.9	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.6	4.1	5.9	8.0	11.3	14.7	17.4	16.9	15.0	11.7	7.9	6.5
	DBStd	5.38	3.54	3.37	2.65	2.77	2.48	2.63	2.33	2.36	2.41	2.84	3.22	2.95
	HDD10.0	761	168	167	129	74	14	1	0	0	17	78	113	(8)
	HDD18.3	2945	425	399	384	311	217	113	48	57	103	207	313	367
	CDD10.0	894	1	1	3	13	56	141	228	214	150	68	14	4
	CDD18.3	36	0	0	0	0	0	3	17	13	3	0	0	0
	CDH23.3	132	0	0	0	0	1	16	72	41	3	0	0	0
	CDH26.7	16	0	0	0	0	0	1	13	2	0	0	0	0
(14) Wind	WSAvg	3.6	4.4	4.4	4.0	3.3	3.5	3.0	2.9	3.1	3.1	3.5	3.5	4.0
(15) Precipitation	PrecAvg	596	52	40	38	38	45	53	50	59	47	64	57	55
	PrecMax	764	102	87	93	103	116	127	108	122	105	140	103	107
	PrecMin	428	7	8	4	0	5	10	9	6	4	5	15	17
	PrecStd	103	23	21	24	27	28	35	23	31	26	32	25	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	13.2	15.9	18.2	21.1	25.2	28.0	26.2	23.9	20.2	15.8	14.0
		MCWB	10.1	10.5	12.0	12.8	15.0	17.6	19.9	18.7	17.2	16.4	12.8	11.4
	2%	DB	11.9	11.9	12.9	16.1	18.9	23.0	25.1	24.2	22.0	18.1	14.2	12.9
		MCWB	9.3	9.5	9.5	11.8	13.9	17.1	18.5	17.9	16.8	14.6	11.5	10.7
	5%	DB	10.9	10.1	11.2	14.2	17.2	21.2	23.2	23.0	20.8	17.0	13.2	12.0
		MCWB	8.8	7.8	8.2	10.7	13.0	16.2	17.6	17.0	16.2	14.1	10.9	9.8
	10%	DB	9.2	8.9	10.1	12.9	16.0	19.8	22.0	21.8	19.1	16.0	12.8	11.1
		MCWB	7.5	6.9	7.5	9.8	12.0	15.3	16.9	16.7	15.3	13.2	10.6	9.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.9	12.0	14.0	15.6	19.4	20.5	19.9	18.7	16.9	12.9	11.8
		MCDB	12.5	13.2	15.8	17.6	19.6	23.7	25.7	24.2	21.9	19.9	14.9	13.5
	2%	WB	10.0	9.8	10.0	12.3	14.2	17.7	19.4	18.8	17.7	15.2	12.1	10.9
		MCDB	11.7	12.0	12.4	15.1	17.9	22.1	24.5	22.7	20.6	17.4	14.0	12.8
	5%	WB	8.9	8.1	9.0	11.1	13.4	16.5	18.4	18.0	16.8	14.2	11.4	10.1
		MCDB	10.4	9.8	11.4	14.4	16.9	20.6	22.5	21.7	19.5	16.4	13.4	11.7
	10%	WB	7.9	6.8	7.8	9.8	12.4	15.4	17.5	17.3	15.9	13.3	10.7	9.4
		MCDB	9.4	8.3	9.7	12.9	15.5	19.1	21.0	20.5	18.5	15.6	12.7	11.0
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.3	6.0	6.7	7.0	7.3	7.6	7.8	7.1	6.0	5.2	4.5
		MCDBR	5.5	5.9	7.4	9.2	9.0	9.8	9.8	9.7	8.9	6.7	5.2	4.8
		MCWBR	5.5	5.1	5.7	6.1	5.7	5.8	5.5	5.1	5.9	4.9	4.6	4.3
	5% WB	MCDBR	5.2	5.8	7.4	9.0	8.9	9.4	9.1	8.9	7.6	6.5	4.9	4.6
		MCWBR	5.4	5.3	5.9	6.0	5.9	5.9	5.6	5.2	5.7	5.2	4.6	4.4
(40) Clear-Sky Solar Irradiance	taub	0.328	0.346	0.382	0.403	0.409	0.412	0.411	0.401	0.380	0.362	0.334	0.319	(40)
	taud	2.363	2.377	2.276	2.246	2.281	2.293	2.309	2.353	2.405	2.425	2.404	2.352	(41)
	Ebn at Noon	664	765	802	830	844	843	837	824	799	736	652	605	(42)
	Edh at Noon	62	80	106	123	125	125	121	110	93	76	59	53	(43)
(44) All-Sky Solar Radiation	RadAvg	0.75	1.42	2.54	4.02	4.84	5.22	4.97	4.17	3.13	1.81	0.92	0.59	(44)
	RadStd	0.07	0.21	0.29	0.41	0.36	0.41	0.49	0.31	0.26	0.17	0.10	0.07	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page